

Computational Geometry Foundations

1 What Is Computational Geometry?

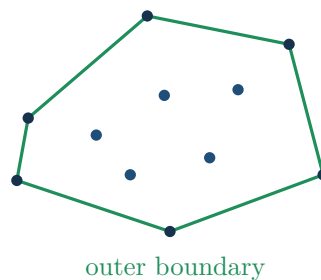
Computational Geometry

Computational geometry studies algorithms for geometric objects such as points, line segments, vectors, polygons, angles, areas, and convex hulls.

The central geometric problem in these notes is:

Convex Hull Motivation

Given a set of points in the plane, find the outer boundary that encloses all of them. This outer boundary is called the **convex hull**.



2 Point Notation

A point in the plane is written as an ordered pair:

$$P = (p_x, p_y).$$

For two points,

$$P = (p_x, p_y), \quad Q = (q_x, q_y).$$

Here p_x, q_x are x -coordinates, and p_y, q_y are y -coordinates.

3 Vectors Between Points

Vector from P to Q

The vector from P to Q is obtained by subtracting coordinates:

$$\overrightarrow{PQ} = Q - P = (q_x - p_x, q_y - p_y).$$

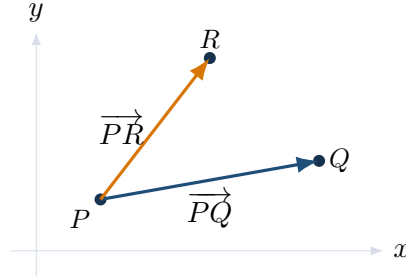
For three points

$$P = (p_x, p_y), \quad Q = (q_x, q_y), \quad R = (r_x, r_y),$$

the two vectors starting at P are

$$\overrightarrow{PQ} = (q_x - p_x, q_y - p_y),$$

$$\overrightarrow{PR} = (r_x - p_x, r_y - p_y).$$



These vectors decide whether the path

$$P \rightarrow Q \rightarrow R$$

turns left, turns right, or stays straight.

4 2D Cross Product

The 2D cross product of vectors

$$\vec{a} = (a_x, a_y), \quad \vec{b} = (b_x, b_y)$$

is the scalar

$$\vec{a} \times \vec{b} = a_x b_y - a_y b_x.$$

For points P, Q, R , use

$$\vec{a} = \overrightarrow{PQ}, \quad \vec{b} = \overrightarrow{PR}.$$

5 Orientation Formula

Orientation Test

For

$$P = (p_x, p_y), \quad Q = (q_x, q_y), \quad R = (r_x, r_y),$$

define

$$\text{orient}(P, Q, R) = (q_x - p_x)(r_y - p_y) - (q_y - p_y)(r_x - p_x).$$

Equivalently,

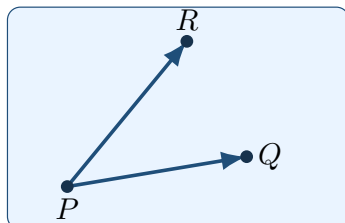
$$\text{orient}(P, Q, R) = \overrightarrow{PQ} \times \overrightarrow{PR}.$$

In determinant form:

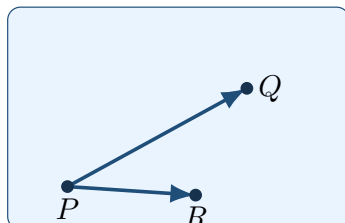
$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} q_x - p_x & q_y - p_y \\ r_x - p_x & r_y - p_y \end{vmatrix}.$$

6 Meaning of the Sign

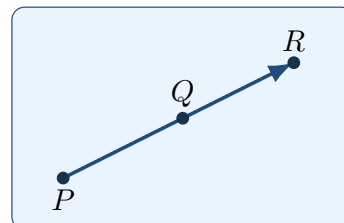
Value of $\text{orient}(P, Q, R)$	Meaning
Positive	Left turn, counterclockwise orientation.
Negative	Right turn, clockwise orientation.
Zero	Collinear points.



$\text{orient} > 0$: left turn



$\text{orient} < 0$: right turn



$\text{orient} = 0$: collinear

7 Worked Examples

Each example uses the same two-step method:

$$\overrightarrow{PQ} = Q - P, \quad \overrightarrow{PR} = R - P,$$

then

$$\text{orient}(P, Q, R) = \overrightarrow{PQ} \times \overrightarrow{PR}.$$

7.1 Left Turn

Example 1

$$P = (1, 1), \quad Q = (4, 2), \quad R = (2, 5).$$

$$\overrightarrow{PQ} = (4 - 1, 2 - 1) = (3, 1),$$

$$\overrightarrow{PR} = (2 - 1, 5 - 1) = (1, 4).$$

Now compute:

$$\text{orient}(P, Q, R) = 3 \cdot 4 - 1 \cdot 1 = 12 - 1 = 11.$$

Since

$$11 > 0,$$

the movement from P to Q to R is counterclockwise.

Result

$$\text{orient}(P, Q, R) > 0 \Rightarrow \text{left turn.}$$

7.2 Right Turn

Example 2

$$P = (1, 1), \quad Q = (2, 5), \quad R = (4, 2).$$

$$\overrightarrow{PQ} = (2 - 1, 5 - 1) = (1, 4),$$

$$\overrightarrow{PR} = (4 - 1, 2 - 1) = (3, 1).$$

$$\text{orient}(P, Q, R) = 1 \cdot 1 - 4 \cdot 3 = 1 - 12 = -11.$$

Since

$$-11 < 0,$$

the movement from P to Q to R is clockwise.

Result

$$\text{orient}(P, Q, R) < 0 \quad \Rightarrow \quad \text{right turn.}$$

7.3 Collinear Points

Example 3

$$P = (0, 0), \quad Q = (2, 2), \quad R = (4, 4).$$

$$\overrightarrow{PQ} = (2, 2), \quad \overrightarrow{PR} = (4, 4).$$

$$\text{orient}(P, Q, R) = 2 \cdot 4 - 2 \cdot 4 = 8 - 8 = 0.$$

Result

$$\text{orient}(P, Q, R) = 0 \quad \Rightarrow \quad P, Q, R \text{ are collinear.}$$

7.4 Quick Comparison

Case	Points	orient	Meaning
Left turn	$P = (1, 1), Q = (4, 2), R = (2, 5)$	11	counterclockwise
Right turn	$P = (1, 1), Q = (2, 5), R = (4, 2)$	-11	clockwise
Collinear	$P = (0, 0), Q = (2, 2), R = (4, 4)$	0	same straight line

8 Why Orientation Is Useful

The orientation test is preferred over angle measurement because it:

- avoids trigonometric functions,
- is fast to compute,
- is exact for integer coordinates when integer arithmetic is used,
- is the basic test behind convex hull algorithms.

Implementation Warning

For large integer coordinates, the products

$$(q_x - p_x)(r_y - p_y) \quad \text{and} \quad (q_y - p_y)(r_x - p_x)$$

may overflow fixed-size integer types. Use a sufficiently large numeric type.

9 Final Review

Foundations Summary

$P = (p_x, p_y)$ represents a point in the plane.

$\overrightarrow{PQ} = (q_x - p_x, q_y - p_y)$.

$\text{orient}(P, Q, R) = \overrightarrow{PQ} \times \overrightarrow{PR}$.

$\text{orient} > 0$: left turn; $\text{orient} < 0$: right turn; $\text{orient} = 0$: collinear.